In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

In [2]:
```
df = pd.read_csv("train.csv")
```

In [3]:
```
df.head()
```

Out[3]:

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Co
0	1	CA-2017-152156	08/11/2017	11/11/2017	Second Class	CG-12520	Claire Gute	Consumer	U
1	2	CA-2017-152156	08/11/2017	11/11/2017	Second Class	CG-12520	Claire Gute	Consumer	U
2	3	CA-2017-138688	12/06/2017	16/06/2017	Second Class	DV-13045	Darrin Van Huff	Corporate	U
3	4	US-2016-108966	11/10/2016	18/10/2016	Standard Class	SO-20335	Sean O'Donnell	Consumer	U
4	5	US-2016-108966	11/10/2016	18/10/2016	Standard Class	SO-20335	Sean O'Donnell	Consumer	U

In [4]:
```
df.columns = df.columns.str.strip()
```

In [5]:
```
df.info()
df.describe()
df.isnull().sum()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9800 entries, 0 to 9799
Data columns (total 18 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Row ID         9800 non-null   int64
 1   Order ID       9800 non-null   object
 2   Order Date     9800 non-null   object
 3   Ship Date      9800 non-null   object
 4   Ship Mode      9800 non-null   object
 5   Customer ID    9800 non-null   object
 6   Customer Name  9800 non-null   object
 7   Segment        9800 non-null   object
 8   Country        9800 non-null   object
 9   City           9800 non-null   object
 10  State          9800 non-null   object
 11  Postal Code    9789 non-null   float64
 12  Region         9800 non-null   object
 13  Product ID     9800 non-null   object
 14  Category       9800 non-null   object
 15  Sub-Category   9800 non-null   object
 16  Product Name   9800 non-null   object
 17  Sales          9800 non-null   float64
dtypes: float64(2), int64(1), object(15)
memory usage: 1.3+ MB
```

Out[5]:

```
Row ID            0
Order ID          0
Order Date        0
Ship Date         0
Ship Mode         0
Customer ID       0
Customer Name     0
Segment           0
Country           0
City              0
State             0
Postal Code      11
Region            0
Product ID        0
Category          0
Sub-Category      0
Product Name      0
Sales             0
dtype: int64
```

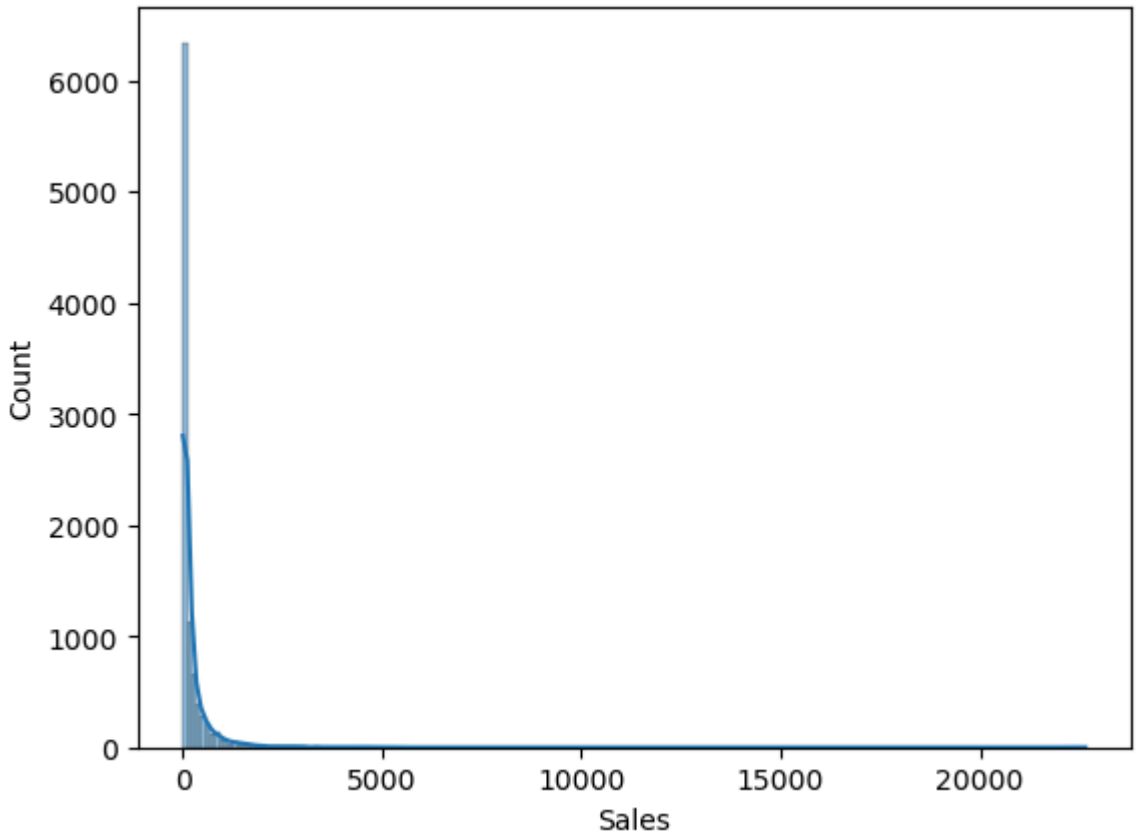
In [7]:

```
df['Order Date'] = pd.to_datetime(df['Order Date'], dayfirst=True)
df['Ship Date'] = pd.to_datetime(df['Ship Date'], dayfirst=True)
```

In [8]:

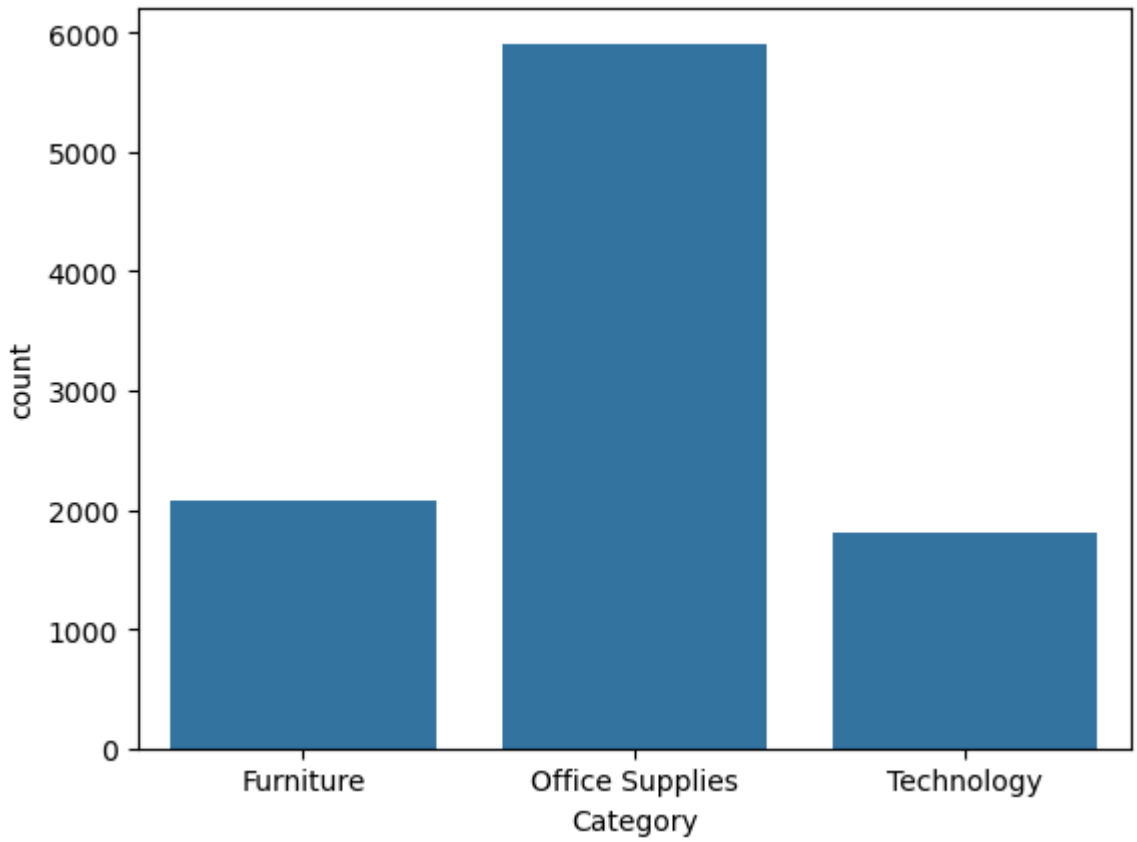
```
import seaborn as sns
import matplotlib.pyplot as plt

sns.histplot(df['Sales'], kde=True)
plt.show()
```



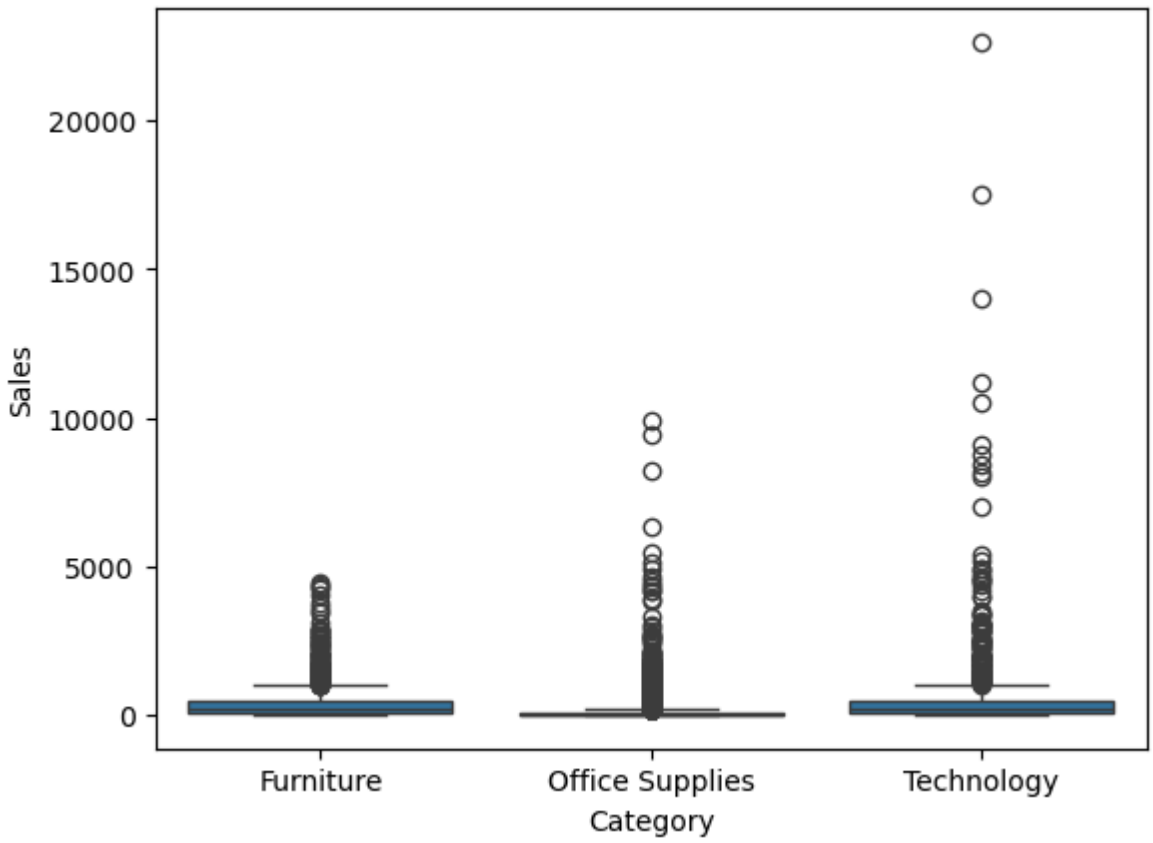
Observation: Sales are right-skewed, meaning most orders have low sales while a few orders have very high values.

In [9]:
```
sns.countplot(x='Category', data=df)
plt.show()
```



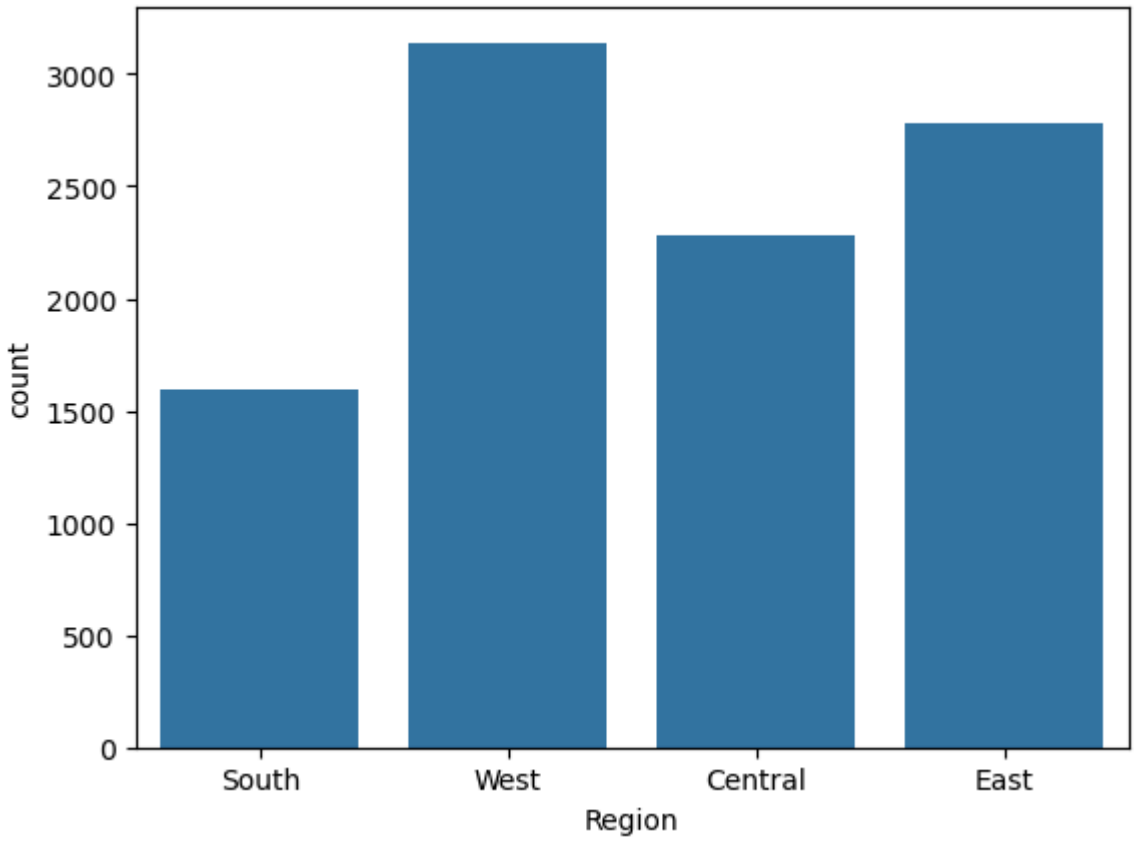
Observation: Office Supplies category has the highest number of orders.

In [10]:
```
sns.boxplot(x='Category', y='Sales', data=df)
plt.show()
```



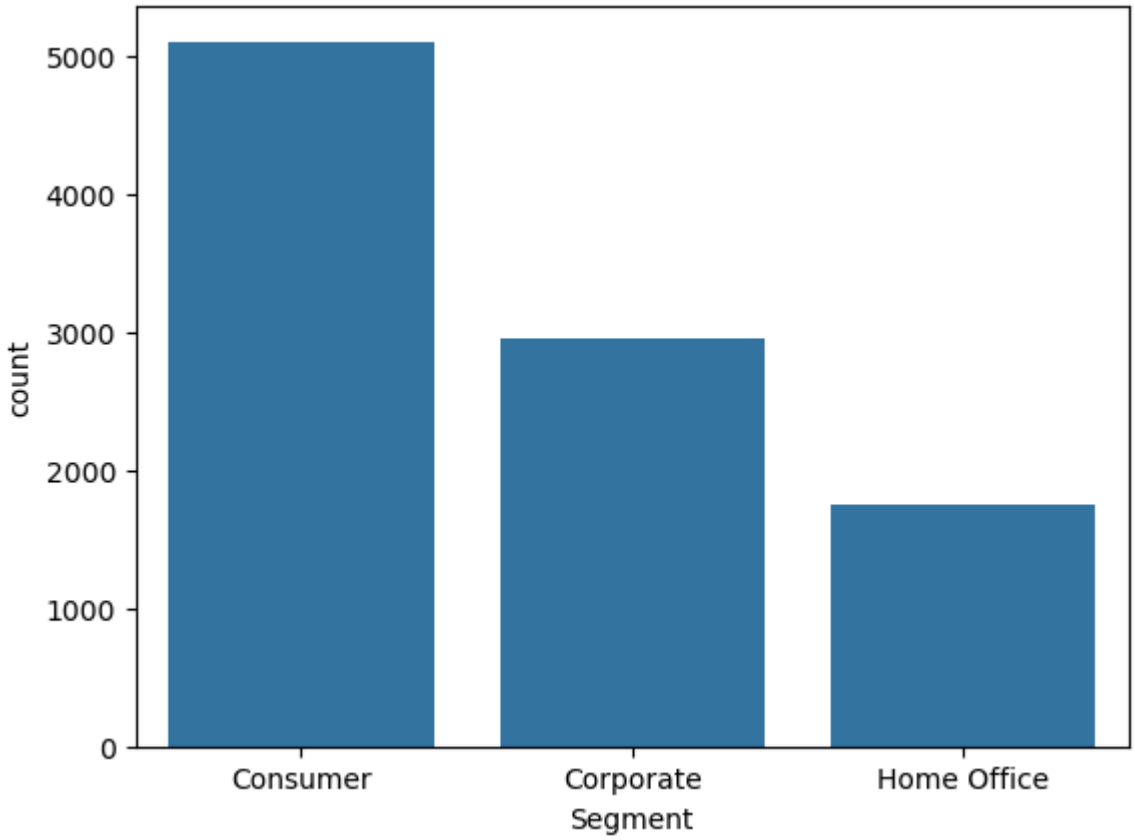
Observation: Technology category has higher sales values and more variation.

In [11]:
```
sns.countplot(x='Region', data=df)
plt.show()
```



Observation: Orders are distributed across regions with some regions having higher activity.

In [12]:
```
sns.countplot(x='Segment', data=df)
plt.show()
```



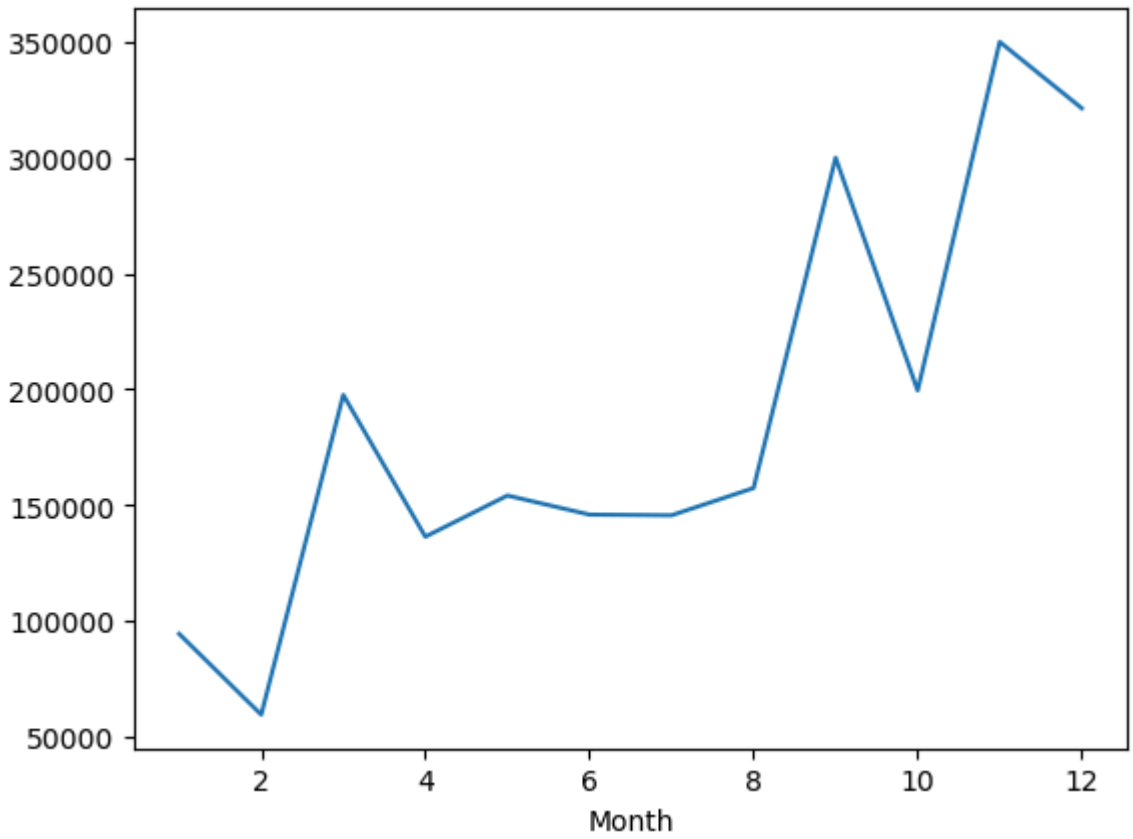
Observation: Consumer segment contributes the highest number of orders.

In [13]:
```
df.groupby('State')['Sales'].sum().sort_values(ascending=False).head()
```

Out[13]:
```
State
California      446306.4635
New York        306361.1470
Texas           168572.5322
Washington      135206.8500
Pennsylvania    116276.6500
Name: Sales, dtype: float64
```

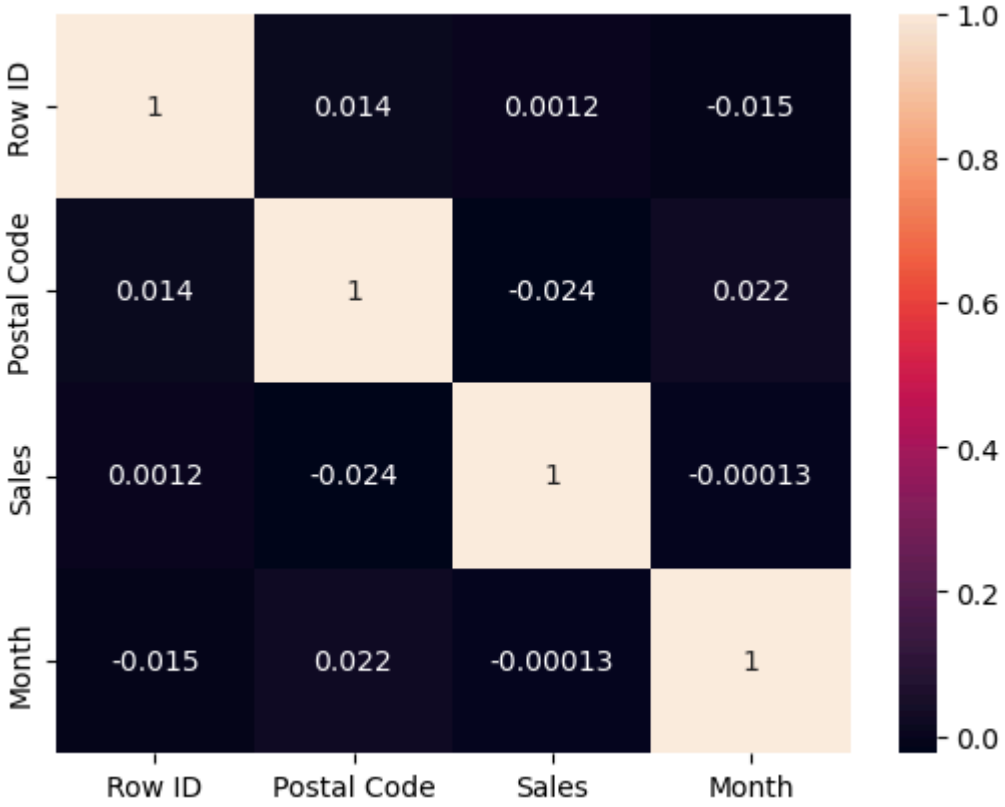
In [14]:
```
df['Month'] = df['Order Date'].dt.month

df.groupby('Month')['Sales'].sum().plot()
plt.show()
```



Observation: Sales vary across months, indicating seasonal trends.

In [15]:
```
sns.heatmap(df.corr(numeric_only=True), annot=True)
plt.show()
```



Observation: Numerical variables show weak correlation.

Conclusion:

- Sales are unevenly distributed with few high-value transactions.
- Technology category generates higher revenue.
- Consumer segment dominates orders.
- Some states and regions contribute more to sales.
- Sales show variation over time indicating trends.

In [ ]: